

# PRODUCT REQUIREMENT DOCUMENT (PRD)

## Multi-Source AI Assistant: LangChain & LangGraph Integration

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### 1. Executive Overview & Problem Statement

Our Multi-Source Chatbot system receives diverse incoming requests across customer care, business analytics, knowledge retrieval, and system actions. Historically, query routing and security boundaries were managed deterministically via Native Router v2.2. We evaluated LangChain and LangGraph to determine their fitness for production routing, safety, and agentic workflows.

### 2. Architecture & Implementation Summary

- > python-langchain-service (Port 8003): LangChain LLM bindings (ChatOpenAI) + LangGraph stateful cyclic agent graph (MAX\_LOOP\_DEPTH = 5).
- > Deterministic Security Gate: Strictly intercepts LLM tool calls. Enforces read-only tools and auto-injects workspace\_id/customer\_id.
- > Laravel Orchestration: Integrated via LangChainRouter.php capturing tool\_calls, iterations, and final\_answer.

### 3. Empirical Benchmark Results (38 Evaluation Queries)

| Metric              | Native v2.2 | LangGraph L1 | LangGraph L1.1 | LangGraph L2  |
|---------------------|-------------|--------------|----------------|---------------|
| Route Accuracy      | 94.74%      | 84.21%       | 73.68%         | 52.63%        |
| Security Accuracy   | 97.37%      | 94.74%       | 92.11%         | 86.84%        |
| Mutation Recall     | 100.00%     | 100.00%      | 100.00%        | 50.00%        |
| Analytics Recall    | 100.00%     | 100.00%      | 70.00%         | 50.00%        |
| Avg Latency         | ~1.00 s     | ~0.98 s      | ~0.99 s        | ~2.24 s       |
| P95 Latency         | ~1.26 s     | ~1.31 s      | ~1.33 s        | ~4.47 s       |
| Tool Calls / Query  | -           | -            | -              | 0.45 (17 tot) |
| Agent Loops / Query | 1.0         | 1.0          | 1.0            | 1.39          |

### 4. Key Scientific & Engineering Insights

- > Routing vs Problem Solving: Giving the LLM tool autonomy (bind\_tools) shifts its behavior from strict classification to open-ended conversation, dropping routing accuracy from 84.21% to 52.63%.
- > Prompt Fragility (L1.1): Adding natural language rules caused over-generalization of ambiguous nouns, reducing accuracy to 73.68%.
- > Latency Overhead: Agentic reasoning loops incur a 2.2x latency tax (~2.24s avg, ~4.47s P95) vs single-pass routing (~1.0s).

### 5. Final Architecture: Two-Tier Production Pattern

- > Tier 1 - First-Hop Control (Native Router v2.2): Primary ingress gateway: Fast (1.0s), deterministic, 94.7% accuracy, zero prompt drift.
- > Tier 2 - Downstream Task Worker (LangGraph Agent): Downstream execution node for multi-turn workflows requiring autonomous multi-tool data aggregation.